Jacob's Projects - HDB financing

Early HDB loans lost to any bank charging under 1.4 points

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Abstract

The HDB concessionary loan has been fixed at 2.6 per cent since 1999 while bank rates moved with the market. The realised numbers show that a bank had to charge under 1.4 points over SORA for a 2010-2015 borrower to have done better, but three of the eight predictions failed outright.

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JEL: G21, R31, E43

The HDB concessionary loan has sat at 2.6 per cent since 1999. Singapore bank rates ran well below that for most of the 2010s, then rose above it from late 2022 to early 2025. By loan start year, the gap was not close.

The HDB concessionary rate is pegged 0.1 percentage points above the CPF Ordinary Account rate, and the Ordinary Account has sat at its legislated 2.5 per cent floor in every quarter a primary CPF Board document publishes, from 1999 to 2026 (Central Provident Fund Board 2026b). The HDB rate has therefore been 2.6 per cent, flat, for the entire period this piece covers (Housing and Development Board 2026).

Compounded 3-month SORA, the public benchmark Singapore mortgages have priced off since 2020, moved a great deal over the same window: far below 2.6 per cent for more than a decade, above it for the 28 months from November 2022 to February 2025, and back below it from March 2025 (Monetary Authority of Singapore 2026b). A borrower who took the HDB loan in, say, 2011 was paying above the benchmark for years before the benchmark ever caught up.

No Singapore authority publishes what a bank actually charged on top of that benchmark, so this piece solves for it instead of assuming it, cohort by cohort. Eight results were scored against wording sealed before any rate data was read. Five passed and three failed. This is a historical comparison by loan start year, past tense and cohort-level throughout, and not advice: Jacob is not licensed under the Financial Advisers Act, and none of this is tailored to any individual borrower's own loan.

1. No published rate to compare against, so the margin is solved for

MAS's own Monthly Statistical Bulletin, Table III.3, lists eleven Singapore interest rate series and not one of them is a bank's housing loan rate (Monetary Authority of Singapore 2026a). The only housing loan rate anywhere in the table belongs to finance companies, smaller deposit-taking lenders funded more expensively than banks, and it runs only to June 2022, before the rate rise the whole piece turns on. It is kept as a labelled side comparison and never treated as a bank proxy.

So the design solves for the unknown instead of guessing it. For each loan start year from 2010 to 2025, at a fixed S$400,000 principal over 25 years, the calculation finds the constant margin over compounded 3-month SORA that would have left a bank borrower's cumulative interest, from loan

start to the last data month, exactly level with an HDB borrower's on 2.6 per cent. That margin is the realised break-even spread. It contains no assumption about future rates, because it is measured only over months that actually happened.[1]

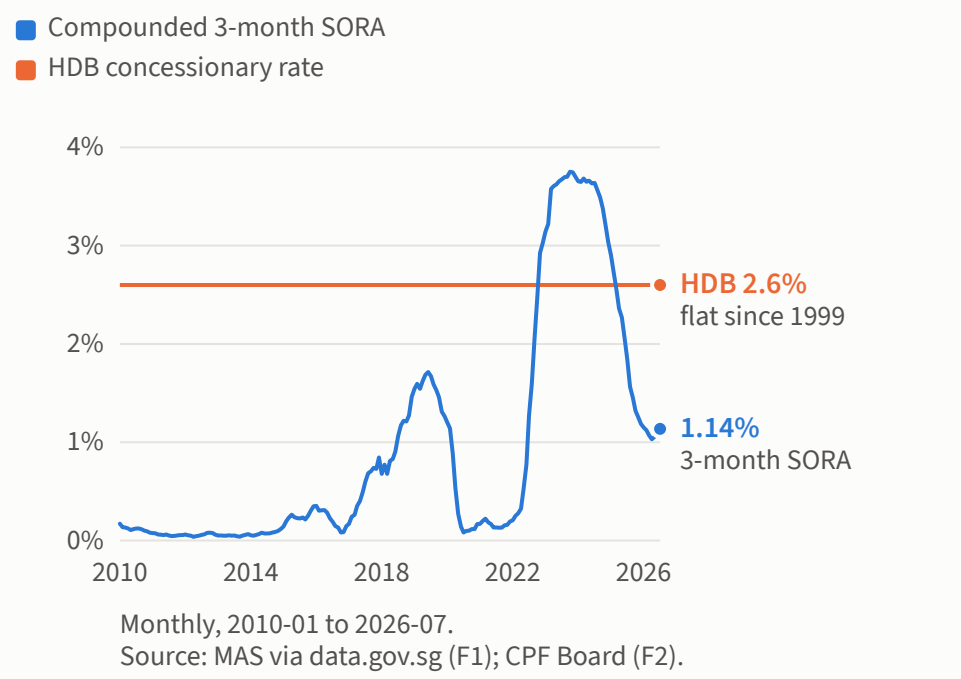


Figure 1: Compounded 3-month SORA against the flat HDB rate, 2010 to 2026. SORA runs far below 2.6 per cent for more than a decade, rises above it from November 2022 to February 2025, and falls back below it from March 2025.

2. Before 2016, a bank had to charge over 1.4 points to lose

A loan that started in 2010 needed a bank charging between 1.89 and 2.05 percentage points over compounded 3-month SORA, depending on the reset interval assumed, to come out level with the HDB route by the last data month. That is the widest gap of any cohort at every one of the four reset intervals tested.

Every one of the six start years from 2010 to 2015 cleared 1.0 point at every tested reset interval, without exception. The lowest figure anywhere in that block was 1.43 points, at a 2015 start on the 3-month reset; at the base 36-month reset the lowest was 1.64 points, at a 2014 start.

> **For 2010-2015 starts a pass is weak evidence and a fail is strong evidence, because the benchmark substitution favours a pass.**

That caveat is not decoration. Before 2020, Singapore mortgages priced off SIBOR, not SORA, and SIBOR ran above SORA. Measuring the margin over SORA for those years therefore overstates the margin a bank would actually have needed, which pushes the result toward exactly what was found. A fail for those cohorts would have survived that bias; a pass has to be read with it stated alongside, which is why it travels with the number rather than sitting in a footnote.

By 2023 the picture had flipped entirely. That is the first start year whose realised break-even spread came in negative at every tested reset interval, ranging from −0.19 to −0.44 points depending on the

[1] **Why the reset interval is tested four ways.** The base case reprices the bank loan to the benchmark every three years, which the sealed thesis calls "the closest thing to a neutral assumption" rather than a fact about any specific mortgage package, since no contractual reset schedule is publicly sourced either. Every result in this piece is also recomputed at 24, 12 and 3-month resets, so a reader can see how much of a finding depends on that one assumption and how much does not.

reset. A negative spread means the HDB loan beat compounded SORA itself, before any bank margin was added on top.

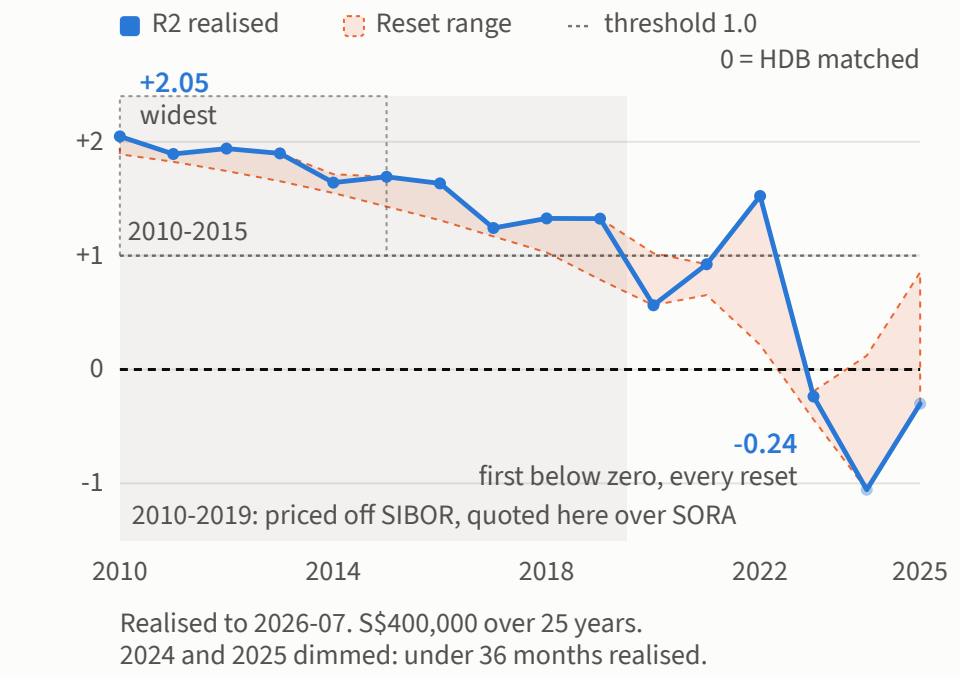


Figure 2: Realised break-even spread by start year, with the band across 3, 12, 24 and 36-month resets. The 1.0 threshold and the zero line are both marked; the 2010-2015 span is bracketed, and pre-2020 starts are shaded to flag the SIBOR substitution.

3. Three predictions that did not survive the data

Eight results in all were scored against wording sealed before any rate data was read: T1a, T1b, T2a, T2b, T3, T4a, T4b and S1. Five passed and three failed, and none of the three failures is buried in a footnote.

Table 1: The eight sealed predictions and their outcomes, as stated on the site's scorecard (3 October 2026).

Sealed prediction	Outcome	Result
2010-2015 starts needed a bigger bank margin to break even than 2020-2023 starts	Held	lowest 2010-2015 figure 1.64; highest 2020-2023 figure 1.52
Every 2010-2015 start needed more than 1.0 point over the benchmark for a bank to lose (the headline)	Held	1.64 to 2.05; weak evidence, because the one bias in the design favours it
No 2010-2015 bank advantage flipped before 2022 at margins up to 0.75	Held	none did
Every 2020-2023 start would flip to an HDB advantage within three years	Failed	15 of 24 cases did not, 10 of them never flipped at all
Refinancing whenever it paid would matter most for 2010-2015 starts and hardly at all for 2022-2025	Failed	the reverse, 0.30 against 1.26 points
Switching from HDB to a bank in 2021 would leave every borrower worse off	Failed	31 of 66 cases worse, 35 not
Switching early, in 2012 or 2015, would pay for 2010-2011 starts at margins up to 0.75	Held	12 of 12 did
The finance-company housing rate stayed above 2.6 per cent in every published month	Held	150 of 150 months

Two of the passes sit beyond the headline. T2a predicted that no 2010-2015 start would cross over to an HDB advantage before 2022 at spreads up to 0.75 points, and none did. T4b predicted that an early switch out of the HDB loan would pay, and all twelve of those cases came out ahead.

The prediction that every 2020-2023 start would flip to an HDB advantage within three years, at every spread tested, failed. At the base 36-month reset, 15 of 24 start-year-by-spread cells missed that window, and 10 of those never crossed at all inside the data available.

The sealed prediction about refinancing had two halves, and both failed. It compared R3, refinancing whenever it paid, scored with hindsight as an upper bound, against R2, repricing on schedule every 36 months at the same spread. It predicted the gap between them would be largest for 2010-2015 starts and below 0.2 points for 2022-2025 starts. The result was the reverse: the largest gap sat with 2022-2025 starts at 1.260 points, against 0.296 points for 2010-2015.

The prediction that switching from the HDB loan to a bank loan in 2021 would leave every applicable cohort worse off than staying, at every spread tested, also failed. Of 66 start-year-by-spread cases, 31 came in worse. Thirty-five did not.

4. The borrower who switched and could not switch back

An HDB borrower can refinance to a bank loan; the reverse move is not available. That asymmetry was modelled directly on a 2010 start, switching once in 2021 and then repricing on schedule afterward with no further shopping around, the same rule the 2021-switch prediction above was tested against.

At an assumed margin of 1.25 points over SORA, that switch cost $128,311 in cumulative interest by the last data month, against $125,576 for staying on the HDB loan throughout, a difference of $2,735. At an assumed margin of 0.25 points, the identical switch in the identical year cost $116,183, some $9,393 less than staying. Whether the 2021 switch helped or hurt the same borrower depended entirely on the margin assumed, which is exactly why a prediction that it would hurt everyone, at every margin, could not hold.

Switching earlier told a cleaner story. Every one of twelve cases where a 2010 or 2011 start switched in 2012 or 2015, at a margin of 0.75 points or below, came out ahead of staying on the HDB loan.[2]

5. What would prove this wrong

A published bank mortgage margin over SORA or SIBOR for 2010-2015, if one is ever released, that came in below roughly 1.4 points would overturn the realised advantage reported for those cohorts, because that is the floor every one of them cleared here. No such figure exists in any source read for this piece: the 2025 Financial Stability Review and the November 2021 Information Paper on Residential Mortgage Pricing were both read for one; neither publishes a level or a range (Monetary Authority of Singapore 2025, 2021).

A sourced official SIBOR history would matter more narrowly: it could move the 2010-2015 range without reversing the finding that the range clears 1.0.

6. What this does not explain

No 2010-2025 start completes a full 25-year tenure inside the data available, so every number here is realised to the last data month rather than a full-tenure outcome, and no forward rate path is used to fill the gap.

What a bank actually charged is still not observable. The margin reported is solved for, not read from a bill, because no Singapore authority publishes one.

Fixed-rate and teaser packages are approximated as a constant rate set at inception, which is not how they behaved month to month; no primary source gives that history either.

[2] **A finance-company rate that never fell below HDB, and stops in June 2022.** The only housing loan rate in MAS Table III.3, from finance companies rather than banks, charged more than the HDB rate in every one of the 150 months it is published inside the 2010-2025 window, without exception, ranging from 2.81 to 5.56 per cent. It is an upper bound on what a costlier class of lender charged, not an estimate of a bank rate, and the series stops in June 2022, at the start of the period the whole piece turns on.

Where an individual borrower's margin sat inside their cohort's range is not something this design can place.

7. What to take from this

The expected direction held up: the earliest cohorts carried the largest gap, and it stayed large even after allowing for the one bias that could have inflated it. Less expected was how cleanly the picture reversed by 2023, when the same fixed rate began beating the benchmark outright.

That three of the mechanics predictions failed matters more than any one of them sounds. Crossover timing, which cohorts refinancing helped most, and whether the 2021 switch hurt everyone all turned on assumptions the public rate path does not settle. An official SIBOR history is the one open item that could turn the SORA-for-SIBOR bias from a stated direction into a measured number.

Data and code

Retrieved 17 September 2026 unless stated otherwise. Full directory, raw files, scripts, checksums and the number manifest in the hdb-affordability repository, subdirectory `financing/` (Ong 2026). CPF Board rate announcements for Q1 2016, Q1 2025 and Q3 2026 were also read (Central Provident Fund Board 2026a).

Not obtained: an official SIBOR history, which would measure the pre-2020 substitution rather than only sign it; and any primary source publishing a bank's housing loan margin over a benchmark.

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